Aksel Kretsinger-Walters, adk2164
adk2164@columbia.edu

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MEMORYLLM

Motivation. Large language models encode knowledge implicitly in their parameters and operate within fixed context windows, limiting their ability to maintain persistent state or accumulate long-term memory across interactions.

Core Idea. MEMORYLLM introduces explicit, learnable memory slots that persist across inputs, allowing the model to maintain and update a structured internal state separate from transient token representations.

Memory Mechanism. A fixed set of memory vectors is jointly processed with input tokens via attention, and read/write operations update these memory slots over time, enabling continuous state evolution beyond a single context window.

Separation of Roles. Token embeddings handle local processing and immediate context, while memory vectors store longer-term information, preventing interference between short-term computation and persistent knowledge.

Persistent State. Unlike segment-level recurrence mechanisms, MEMORYLLM maintains memory across sessions or extended interactions, supporting multi-step reasoning, tool use, and stateful behavior.

Relation to Continual Learning. The architecture resembles continual learning systems in that it accumulates information incrementally in explicit memory rather than solely modifying model parameters, thereby reducing destructive interference.

Comparison to Recurrent Memory Transformer. Where RMT focuses on compressing long sequences through segment recurrence, MEMORYLLM emphasizes long-lived, persistent memory suitable for agent-like applications and interactive settings.

Key Takeaway. MEMORYLLM reframes large language models as stateful systems with explicit memory substrates, separating transient attention-based computation from persistent storage and enabling scalable, long-term reasoning.